

Label agreement between corpora

Results

The model is at 96% of the ceiling the labels can measure. Two independent judge models labelled the same **4,708 datax documents**, and across the 20 PII tags with support ≥ 30 they agree with each other at **macro F1 0.5138** (support-weighted **0.4686**). per label_v4 scores macro F2 **0.4932**.

That is the whole explanation for the plateau. Five model families and 1,000 trials converged to 0.42–0.49 not because the search was inadequate but because **no model can be measured above the reliability of the labels measuring it**, and that reliability is ≈ 0.51 .

Evidence	Value
Target as written	0.90
Champion's bootstrap CI upper bound (optimistic)	0.5892
Inter-judge macro F1 (model-independent)	0.5138
per label_v4 today	0.4932
Verdict	unlikely

Recorded in `feasibility_agreement.json`; the loop is now `target_infeasible` with `renegotiated_target = 0.5138`.

The disagreement is mostly taxonomic, not perceptual

The per-tag numbers looked alarming until the pattern showed itself. The two judges are not disagreeing about what is in the documents. They are disagreeing about **which tag to file it under**.

Tag	Judge A +	Judge B +	F1
full_name	455	1,809	0.3993
given_name	1,513	261	0.2864
family_name	1,547	312	0.3141
middle_name	329	3	0.0181
collapsed to one NAME class	1,878	1,884	0.9920

Judge A decomposes a person's name into given/family/middle. Judge B records `full_name`. Both see the same person: they agree on **name presence at F1 0.9920, Cohen's κ 0.9867** — 1,866 of $\sim 1,880$ documents. Split across four tags, that near-perfect agreement reads as F1 0.018–0.40, and the fixed evaluator scores the difference as *model error*.

`sensitive_pii_full_name` is the project's **highest-support priority tag** (60,902). Its measured ceiling is 0.399. That single taxonomy choice is the largest measurement artifact in the run.

Address shows the same effect more weakly — address 0.1286 and street_number_and_name 0.6953 individually, **0.7396** collapsed — so part is taxonomic and part is genuine disagreement about what counts as an address.

Format-anchored tags need no collapsing and already agree well: email 0.9095, phone_number 0.8939, age 0.8515. Semantically-defined tags are where the reliability goes: employment_status 0.0440, country_of_residence 0.3448, zip_code 0.5661, medical_condition 0.5682.

What collapsing the taxonomy would buy

Merging the four name tags into one and the two street-address tags into one — changing nothing else:

Taxonomy	Macro F1 ceiling	Support-weighted
Current (20 tags)	0.5138	0.4686
Collapsed (16 classes)	0.6354	0.7879

The weighted figure nearly doubles because NAME is the largest class by support. **This is a change to the evaluator, so it is not something to do quietly** — it needs explicit approval and a re-run of every arm under the corrected definition, exactly as widening a policy mid-run would.

Cross-corpus: the corpora also disagree with each other

The eight corpora share no documents, so inter-annotator agreement is undefined across them. What *is* measurable: hold the model fixed and ask what score threshold each corpus demands to reach 0.90 recall on the same tag. If they mean the same thing by a tag, the thresholds should be similar.

Tag	Corpora	Threshold spread	Prevalence spread
medical_record_number_mrn	3	3.8×	1.5×
full_name	6	3.1×	2.6×
credit_card_number	4	2.9×	18.4×
password	5	2.1×	29.0×
address	7	2.0×	17.0×
military_identification_number	3	1.5×	167.7×
passport_number	4	1.2×	36.8×
iban	2	1.1×	3.1×

Threshold spread is the more trustworthy column — prevalence can differ for honest compositional reasons (a medical corpus really does hold more MRNs), whereas a 3.8× difference in the score level at which the *same model* finds a corpus's positives means those positives look materially different. MRN, full_name and credit_card are where the corpora least agree about what the tag means.

This is the mechanism behind the 0.63 / 0.23 spread on openpii versus betterdataai_ner_silver noted in the previous report: the same model, the same tags, different definitions.

TL;DR

- **Two independent judges over the same 4,708 documents agree at macro F1 0.5138.** The champion scores 0.4932 — **96% of the ceiling**. The 0.90 target was never reachable, and this evidence is model-independent, so it does not depend on any arm's defects.
- **Most of the disagreement is taxonomic.** The judges agree on whether a name is present at **F1 0.9920 / κ 0.9867**; split across full_name / given_name / family_name / middle_name that reads as 0.018–0.40. The evaluator scores that bookkeeping difference as model error, on the highest-support tag in the project.
- **Collapsing names and street-address raises the ceiling to macro 0.6354 / weighted 0.7879** — the single largest lever available, and roughly 3× more headroom than the per-label threshold work bought. It is an evaluator change and needs sign-off plus a full re-run.
- Format-anchored tags are fine: email 0.909, phone 0.894, age 0.852. Semantic tags are where reliability collapses: employment_status 0.044.
- Loop recorded as target_infeasible, renegotiated_target = 0.5138.

Model scoreboard

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
current_rules	4000_datax-dualjudge-evalset-1.32k					
current_rules	6589_govdocs2-dualjudge-eval20-3.53k					
current_rules	ai4privacy_pii_masking_eval_10k					
current_rules	betterdataai_ner_silver_eval_10k					
current_rules	nemotron_eval_6k					
current_rules	openpii_pii_eval_38k					
current_rules	pii2_eval_30k					
current_rules	pii_holdout_20k					
hash_sgd	4000_datax-dualjudge-evalset-1.32k					
hash_sgd	6589_govdocs2-dualjudge-eval20-3.53k					
hash_sgd	ai4privacy_pii_masking_eval_10k					
hash_sgd	betterdataai_ner_silver_eval_10k					
hash_sgd	nemotron_eval_6k					
hash_sgd	openpii_pii_eval_38k					
hash_sgd	pii2_eval_30k					
hash_sgd	pii_holdout_20k					

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
hash_sgd_f2	4000_datax-dualjudge-evalset-1.32k					
hash_sgd_f2	6589_govdocs2-dualjudge-eval20-3.53k					
hash_sgd_f2	ai4privacy_pii_masking_eval_10k					
hash_sgd_f2	betterdataai_ner_silver_eval_10k					
hash_sgd_f2	nemotron_eval_6k					
hash_sgd_f2	openpii_pii_eval_38k					
hash_sgd_f2	pii2_eval_30k					
hash_sgd_f2	pii_holdout_20k					
hybrid_priority_001	4000_datax-dualjudge-evalset-1.32k					
hybrid_priority_001	6589_govdocs2-dualjudge-eval20-3.53k					
hybrid_priority_001	ai4privacy_pii_masking_eval_10k					
hybrid_priority_001	betterdataai_ner_silver_eval_10k					
hybrid_priority_001	nemotron_eval_6k					
hybrid_priority_001	openpii_pii_eval_38k					
hybrid_priority_001	pii2_eval_30k					
hybrid_priority_001	pii_holdout_20k					
tfidf_linear	4000_datax-dualjudge-evalset-1.32k					
tfidf_linear	6589_govdocs2-dualjudge-eval20-3.53k					
tfidf_linear	ai4privacy_pii_masking_eval_10k					
tfidf_linear	betterdataai_ner_silver_eval_10k					
tfidf_linear	nemotron_eval_6k					
tfidf_linear	openpii_pii_eval_38k					
tfidf_linear	pii2_eval_30k					
tfidf_linear	pii_holdout_20k					
embeddingbag_asl	4000_datax-dualjudge-evalset-1.32k					
embeddingbag_asl	6589_govdocs2-dualjudge-eval20-3.53k					
embeddingbag_asl	ai4privacy_pii_masking_eval_10k					
embeddingbag_asl	betterdataai_ner_silver_eval_10k					
embeddingbag_asl	nemotron_eval_6k					

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
embeddingbag_asl	openpii_pii_eval_38k					
embeddingbag_asl	pii2_eval_30k					
embeddingbag_asl	pii_holdout_20k					
hybrid_priority	4000_datax-dualjudge-evalset-1.32k					
hybrid_priority	6589_govdocs2-dualjudge-eval20-3.53k					
hybrid_priority	ai4privacy_pii_masking_eval_10k					
hybrid_priority	betterdataai_ner_silver_eval_10k					
hybrid_priority	nemotron_eval_6k					
hybrid_priority	openpii_pii_eval_38k					
hybrid_priority	pii2_eval_30k					
hybrid_priority	pii_holdout_20k					
champion_1k	4000_datax-dualjudge-evalset-1.32k					
champion_1k	6589_govdocs2-dualjudge-eval20-3.53k					
champion_1k	ai4privacy_pii_masking_eval_10k					
champion_1k	betterdataai_ner_silver_eval_10k					
champion_1k	nemotron_eval_6k					
champion_1k	openpii_pii_eval_38k					
champion_1k	pii2_eval_30k					
champion_1k	pii_holdout_20k					
champion_1k_bootstrap	8 holdouts (5 complete for F2)				454.5	2.2
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				984.8	1.0
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				483.5	2.1
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				204.6	4.9
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				146.2	6.8
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				674.0	1.5
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				341.6	2.9
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				175.4	5.7
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				130.7	7.6
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				454.5	2.2

Model	Dataset	F1@ 5	F1@ 3	F1@ 1	Docs/s ec	ms/ doc
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				237.5	4.2
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				147.8	6.8
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				113.3	8.8
perlabel_v4	4000_datax-dualjudge-evalset-1.32k					
perlabel_v4	6589_govdocs2-dualjudge-eval20-3.53k					
perlabel_v4	ai4privacy_pii_masking_eval_10k					
perlabel_v4	betterdataai_ner_silver_eval_10k					
perlabel_v4	nemotron_eval_6k					
perlabel_v4	openpii_pii_eval_38k					
perlabel_v4	pii2_eval_30k					
perlabel_v4	pii_holdout_20k					

Measured 2026-08-25 · n=1,000, n=126,129.